

Request Methods

Concert Websockets API 0.5.0

Terminology

Concert: server softwares for robotic services from installation to operation. It is the name of the project of Fleet managment system(FMS). We manage multi-tasks with multi-robots, assign tasks to appropriate robots, and add detail plans for the tasks. It also helps the robot move to other spaces through interworking with elevators and automatic doors. It also provides an API to access the functions and data of the server for customization.

Job: Job is a goal for performing tasks with the objectives presented by the user to perform the desired services. It describes a workflow that includes time schedule, instructions for a specific task. It's a similar concept with a mission.

Preset: A service that provides an interface for users to easily run pre-written jobs, rather than create jobs every time.

Overview

If you want to implement a custom service, You can implement the business logic using concert websockets API. The functions provided are as follows:

- 1. A preset registered in the Concert can be called to enable the robot to perform a Job.
- 2. To implement a business that requires (semi)real-time monitoring and immediate feedback, it provides events on progress of the job is robot is performing.
- 3. It provides functions to implement batch processing by looking up the current job or using the job's inquiry function based on a specific date.

General Considerations

- When the robot is ready to operate in the Concert, create a **Preset**.
- Issues an **API account** to the Concert's user account to enable the Concert WebSockets API.
- Set "shared" to API account with Concert's user account to enable Preset in API account.
- Job status changes and Robot's instruction performance events are automatically received without a separate subscription.[Server-sent events](#)

Connection details

Connection details:

URL	Scheme	Description
ws://{concert-ip}:20002	WS	

Changelog

2022-06-10 version:0.5.0, compatible concert version: higher 1.8.18

- add Method:req_get_recent_failed
- add Method:req_run_recent_failed

2022-01-11 version:0.4.0

- add Method:req_subscribe_worker
- add feedback-worker-status

2021-12-16 version:0.3.0

- add feedback-instruction-group

2021-10-12 version:0.2.0

- add Method:req_event_trigger

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

2021-07-21 version:0.1.0

- Websockets version 0.1.0

2021-04-30 version:0.0.0

- Websockets draft

Messages

Request

Schema

Name	Type	Description
uuid	string	Required <i>unique identifier</i>
method	string	Required
<i>params</i>	object	Optional

Example of payload

```
{
  "uuid": "this_is_request_id",
  "method": "req_login_api",
  "params": {
    "apikey": "this_is_api_key",
    "secret": "this_is_api_secret"
  }
}
```

Response

Schema

Name	Type	Description
uuid	string	Required
<i>method</i>	string	Required
<i>isSuccess</i>	boolean	Required
<i>error</i>	object	Optional
└ <i>code</i>	string	
└ <i>message</i>	string	
└ <i>detail</i>	object	<i>internal concert-services error</i>
<i>payload</i>	object	Optional

Message Schema

- Request
- Response
- Errors

Terminology

- Action
- Task
- Worker
- Report

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- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

Example of payload

```
{
  "uuid": "this_is_request_id",
  "method": "req_login_api",
  "isSuccess": true
}
```

```
{
  "uuid": "this_is_request_id",
  "method": "res_login_api",
  "isSuccess": false,
  "error": {
    "code": "SAS0002",
    "message": "authentication failed",
    "detail": {
      "code": "0001",
      "message": "validation",
      "reason": "ID or PW isn't matched"
    }
  }
}
```

Errors

code	message	Description
SAP0001	json-parse-error	
SAP0002	validateError	
SAP0003	authentication required	
SAP0004	not supported method	
SAS0001	authentication failed	
SAS0002	authentication failed	
SAS0003	preset-service error occurred	
SAS0004	report-service error occurred	
SAS0005	create-job from preset, failure	
SAS0006	task-service error occurred	
SAS0007	create-job from composer, failure	

Terminology

Action

Action refers to the function executed by the instruction when a worker performs a Job. It defines the name of the function (func_name) and the required parameters (args).

Schema

Name	Type	Description
------	------	-------------

message schema

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Name	Type	Description
func_name	string	
args	array	
└ (Anonymous)	object	
└ key	string	
└ type	string	boolean, string, number, object
└ value	depends on "type"	This field is reserved to assign a value to be used in practice. Leave it as null.
└ (default)	depends on "type"	This field can define the default value.
└ (format)	string	This field can define pattern of the value.

Example of Action

Action define for Move

```
{
  "func_name" : "move",
  "args" : [
    { Enter the name of the registered location as the destination.
      "key" : "destination",
      "type" : "string",
      "value" : null
    },
    { If True, parameters for options that attempt to accurately locate the destination and rotate to the specified position when near the destination.
      "key" : "goal_finishing"
      "type" : "boolean",
      "value" : null
    },
    { Timeout for the goal finishing.
      "key" : "finishing_timeout",
      "type" : "number",
      "value" : null,
    }
  ],
}
```

Task

Task is a group of capabilities that the worker can perform.
For example, tasks such as cleaning, delivery, etc.
The Worker registers it's Task information with the Concert and exports it according to the Job's Task you create.

Schema

Name	Type	Description
id	string	
name	string	
actions	Array	
└ (action)	obejct	action

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Worker

who has the necessary functionality to perform the desired service and usually actually performs a given "task" with the robot.

Schema

Name	Type	Description
id	string	
name	string	
status	string	Worker.status
created_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
updated_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
type_specific	object	Information about workers of a specific type
└ location	object	The registered location information in the Concert.
└ map	string	located position of map's id
└ pose2d	object	located position of map's id
└ x	number	x-coordinate
└ y	number	y-coordinate
└ theta	theta	radian
└ semantic_location	string	located resource(managed by fms)'s id

Worker.status

status	Description
idle	State in which a new job can be assigned, without performing a job.
busy	Status in which job is performed.
offline	Disconnected from the Concert for a long time.
error	A state in which a user's action is required or the job can no longer be performed.

Report

A Report is information generated when a Job is assigned to a Worker, who records the progress of the Job in the Report and reports it to the Concert.

Schema

Name	Type	Description
id	string	
title	string	
worker	string	
status		Report.status
instructions	Array	Instruction to configure the Job's functional workflow. The capabilities included in the registered task will be the constructions that make up the Job. It should have id, next, and contains actions to perform.
└ (instruction)	object	
└ id	string	

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Name	Type	Description
└ next	string	
└ action	object	Action
created_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
assigned_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
started_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
finished_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ

Report.status

status	Description
pending	Waiting for execution, waiting for Walker's request in idle state.
assigned	Report was taken by Worker
running	Worker will perform Report
finished	Report was finished by Worker, normally
cancelled	Report has been finished by the user; the report is no longer valid due to user request.
aborted	Report has been finished by Worker. The worker is unable to excecute instructions.

Instruction.status

status	Description
started	
done	worker finished instruction normally.
failed	worker finished instruction abnormally.

Request Message

req_login

- **description:** Log in through the Concert's user account.
- **return:** Returns success/failure.
- **Note:** When API accounting function is added, it will be deleted.

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ id	string	
└ passwd	string	

Example of payload

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Server-sent events

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request

```
{
  "uuid": "this_is_request_id",
  "method": "req_login",
  "params": {
    "id": "this_is_request_id",
    "passwd": "this_is_user_password"
  }
}
```

response

```
{
  "uuid": "this_is_request_id",
  "method": "res_login",
  "isSuccess": true
}
```

req_login_api

- **description:** Log in through the API account you issued.
- **return:** Returns success/failure.

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ apikey	string	
└ secret	string	

Example of payload

request

```
{
  "uuid": "this_is_request_id",
  "method": "req_login_api",
  "params": {
    "apikey": "this_is_api_key",
    "secret": "this_is_api_secret"
  }
}
```

response

```
{
  "uuid": "this_is_request_id",
  "method": "res_login_api",
  "isSuccess": true
}
```

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- req_get_task_worker
- req_compose_job
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- req_run_recent_failed

Server-sent events

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req_get_preset

- description:** Query information from a particular preset.
- return:** Information from a particular preset is returned.

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ id	string	preset's id
└ detail	boolean	default is false, Recommended use false

Example of payload

request

```
{
  "uuid": "8a4f9193-cba6-4c4a-9846-4229d60f884e",
  "method": "req_get_preset",
  "params": {
    "id": "605c186b3ba1fe0041af0974"
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "8a4f9193-cba6-4c4a-9846-4229d60f884e",
  "method": "res_get_preset",
  "payload": {
    "id": "605c186b3ba1fe0041af0974",
    "title": "cafe-table-1 delivery",
    "description": "cafe-table-1 delivery"
  }
}
```

request

```
{
  "uuid": "8a4f9193-cba6-4c4a-9846-4229d60f884e",
  "method": "req_get_preset",
  "params": {
    "id": "605c186b3ba1fe0041af0974",
    "detail": true
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "8a4f9193-cba6-4c4a-9846-4229d60f884e",
  "method": "res_get_preset",
  "payload": {
```


message schema

- Request
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- req_run_recent_failed

Server-sent events

- feedback_report
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```
"description": "cafe-table-1 delivery",
"_id": "605c186b3ba1fe0041af0974",
"title": "cafe-table-1 delivery",
"owner": "603f4ca45257b8003a627aa8",
"options": {
  "instructions": [
    {
      "id": "c60ebd25bfd3efc2585ba88a16ad7401",
      "action": {
        "args": [
          {
            "key": "destination",
            "type": "string",
            "value": "5f1a3ee9cc6e2a00120d17d7",
            "value_alias": {
              "alias": "cafe_table_1",
              "value": "5f1a3ee9cc6e2a00120d17d7"
            },
            "options": {
              "user_input": false
            },
            "isParamArgs": false
          },
          {
            "key": "goal_finishing",
            "type": "boolean",
            "value": true,
            "value_alias": {
              "alias": "true",
              "value": true
            },
            "options": {
              "user_input": false
            },
            "isParamArgs": false
          },
          {
            "key": "finishing_timeout",
            "type": "number",
            "value": 30,
            "value_alias": {
              "alias": "30s",
              "value": 30
            },
            "options": {
              "max": 300,
              "min": 0,
              "user_input": true
            },
            "isParamArgs": false
          }
        ],
        "func_name": "move"
      },
      "next": null
    }
  ],
  "task": "605c17b65d884aa3ef39a8e0",
  "rules": [
    {
      "rule": "compelWorker",
```

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req_get_presets
req_get_report
req_get_reports
req_get_remain_reports
req_exec_preset
req_cancel_report

req_get_task
req_get_task_worker
req_compose_job
req_event_trigger
req_subscribe_worker
req_get_recent_failed
req_run_recent_failed

Server-sent events

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```
      "args": {
        "workerId": "605c17b50ab6fc0041efe9b8"
      }
    },
    "events": []
  },
  "shared": [],
  "created_at": "2021-03-25T04:58:19.828Z",
  "updated_at": "2021-03-25T05:03:21.518Z",
  "id": "605c186b3ba1fe0041af0974"
}
}
```

req_get_presets

- **description:** Look up the list of available presets.
- **return:** A list of Presets is returned.

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ detail	boolean	default is false, Recommended use false

Example of payload

request

```
{
  "uuid": "a4d01f42-c4d6-4fe1-8aa6-69e6b77fa7f1",
  "method": "req_get_presets",
  "params": {
    "detail": false
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "a4d01f42-c4d6-4fe1-8aa6-69e6b77fa7f1",
  "method": "res_get_presets",
  "payload": [
    {
      "id": "605c195e3ba1fe0041af0976",
      "title": "1st_hall_1 round trip",
      "description": "move to 1st_hall_1 return to charging-station"
    },
    {
      "id": "605c186b3ba1fe0041af0974",
      "title": "cafe-table-1 delivery",
      "description": "cafe-table-1 delivery"
    }
  ]
}
```

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Server-sent events

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```
]
}
```

req_get_report

- description:** Query information from a particular report.
- return:** Information from a particular report is returned. [Report](#)

Name	Type	Description
uuid	string	
method	string	
params	object	
└ id	string	report's id
└ detail	boolean	default is false, Recommended use false

Example of payload

request

```
{
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "req_get_report",
  "params": {
    "id": "60c06d97d5bccb002b9c6940",
    "detail": false
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "res_get_report",
  "payload": {
    "id": "60c06d97d5bccb002b9c6940",
    "title": "task_2 (2 / 100)",
    "worker": "60c0251ed5bccb002b9c68b1",
    "status": "finished",
    "created_at": "2021-06-09T07:28:23.661Z",
    "assigned_at": "2021-06-09T07:28:24.054Z",
    "started_at": "2021-06-09T07:28:24.188Z",
    "finished_at": "2021-06-09T07:30:55.410Z",
    "updated_at": "2021-06-09T07:30:55.429Z"
  }
}
```

request

```
{
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "req_get_report",
  "params": {
    "id": "60c06d97d5bccb002b9c6940",
    "detail": true
  }
}
```

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response

```
{
  "isSuccess": true,
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "res_get_report",
  "payload": {
    "priority": -1,
    "customService": null,
    "conditions": [],
    "_id": "60c06d97d5bccb002b9c6940",
    "title": "task_2 (2 / 100)",
    "job": "60c06d27d5bccb002b9c693c",
    "task": "60c06c6325288b4eb8949f70",
    "worker": "60c0251ed5bccb002b9c68b1",
    "created_at": "2021-06-09T07:28:23.661Z",
    "assigned_at": "2021-06-09T07:28:24.054Z",
    "started_at": "2021-06-09T07:28:24.188Z",
    "finished_at": "2021-06-09T07:30:55.410Z",
    "instructions": [
      {
        "id": "1",
        "action": {
          "fixed": false,
          "args": [
            {
              "type": "string",
              "value": "60c017863d370700402154d1",
              "domain": [],
              "default": null,
              "properties": [],
              "key": "destination",
              "value_alias": {
                "value": "60c017863d370700402154d1",
                "alias": "dy_loc_1"
              }
            }
          ],
        },
      },
      {
        "type": "number",
        "value": -1,
        "domain": [],
        "default": null,
        "properties": [],
        "key": "finishing_timeout",
        "value_alias": {
          "alias": "-1",
          "value": -1
        }
      }
    ],
    "func_name": "move"
  },
  "next": "2",
  "X-Inst-Id": "82f239ae12248f97ca465de278bee439",
  "issuer_type": "user",
  "performer_type": "worker"
},
{
  "id": "2",
  "action": {
    "fixed": false,
```

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```
"args": [
  {
    "type": "string",
    "value": "60c0205d3d370700402154d5",
    "domain": [],
    "default": null,
    "properties": [],
    "key": "destination",
    "value_alias": {
      "value": "60c0205d3d370700402154d5",
      "alias": "dy_loc_5"
    }
  },
  {
    "type": "number",
    "value": -1,
    "domain": [],
    "default": null,
    "properties": [],
    "key": "finishing_timeout",
    "value_alias": {
      "alias": "-1",
      "value": -1
    }
  }
],
"func_name": "move"
},
"next": "3",
"X-Inst-Id": "78d6f07e9c9ce38515ffcc7ea11758ec",
"issuer_type": "user",
"performer_type": "worker"
},
{
  "id": "3",
  "action": {
    "fixed": false,
    "args": [
      {
        "type": "string",
        "value": "60c0179c3d370700402154d3",
        "domain": [],
        "default": null,
        "properties": [],
        "key": "destination",
        "value_alias": {
          "value": "60c0179c3d370700402154d3",
          "alias": "dy_loc_3"
        }
      }
    ]
  },
  {
    "type": "number",
    "value": -1,
    "domain": [],
    "default": null,
    "properties": [],
    "key": "finishing_timeout",
    "value_alias": {
      "alias": "-1",
      "value": -1
    }
  }
}
```

Message Schema

Request

Response

Errors

Terminology

Action

Task

Worker

Report

Request Methods

req_login

req_login_api

req_get_preset

req_get_presets

req_get_report

req_get_reports

req_get_remain_reports

req_exec_preset

req_cancel_report

req_get_task

req_get_task_worker

req_compose_job

req_event_trigger

req_subscribe_worker

req_get_recent_failed

req_run_recent_failed

Server-sent events

feedback_report

feedback_instruction

feedback_instruction_group

feedback_worker_status

```
    },
    "func_name": "move"
  },
  "next": null,
  "X-Inst-Id": "f48f5bb70dfecf34c82213689da5e6a3",
  "issuer_type": "user",
  "performer_type": "worker"
}
],
"status": "finished",
"updated_at": "2021-06-09T07:30:55.429Z",
"result": {
  "status": "success",
  "status_msg": "just finished",
  "current": null,
  "returns": [
    {
      "id": "1",
      "X-Inst-Id": "82f239ae12248f97ca465de278bee439",
      "status": "success",
      "status_msg": "just finished",
      "started_at": "2021-06-09T07:28:24.188Z",
      "finished_at": "2021-06-09T07:29:33.942Z",
      "updated_at": "2021-06-09T07:29:33.952Z"
    },
    {
      "id": "2",
      "X-Inst-Id": "78d6f07e9c9ce38515ffcc7ea11758ec",
      "status": "success",
      "status_msg": "just finished",
      "started_at": "2021-06-09T07:29:33.961Z",
      "finished_at": "2021-06-09T07:30:38.923Z",
      "updated_at": "2021-06-09T07:30:38.928Z"
    },
    {
      "id": "3",
      "X-Inst-Id": "f48f5bb70dfecf34c82213689da5e6a3",
      "status": "success",
      "status_msg": "just finished",
      "started_at": "2021-06-09T07:30:38.934Z",
      "finished_at": "2021-06-09T07:30:55.410Z",
      "updated_at": "2021-06-09T07:30:55.429Z"
    }
  ],
  "ownership_executionTime": 0.024
},
"id": "60c06d97d5bccb002b9c6940"
}
```

req_get_reports

- **description:** Look up the list of reports generated on the yyyyymmdd date based on UTC+0.
- **return:** [Report](#) list is returned.

Name	Type	Description
uuid	string	

Message Schema

- Request
- Response
- Errors

Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

Name	Type	Description
method	string	
params	object	
└─ yyyymmdd	string	format: yyyy-mm-dd
└─ detail	boolean	default is false, Recommended use false

Example of payload

request

```
{
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "req_get_reports",
  "params": {
    "yyyymmdd": "2021-03-26"
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "res_get_reports",
  "payload": [
    {
      "id": "605d649b0ab6fc0041efe9c8",
      "title": "cafe-table-1 delivery",
      "worker": "605c17b50ab6fc0041efe9b8",
      "status": "finished",
      "created_at": "2021-03-26T04:35:39.159Z",
      "assigned_at": "2021-03-26T04:35:43.033Z",
      "started_at": "2021-03-26T04:35:47.971Z",
      "finished_at": "2021-03-26T04:35:48.316Z",
      "updated_at": "2021-03-26T04:35:48.322Z"
    }
  ]
}
```

request

```
{
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "req_get_reports",
  "params": {
    "yyyymmdd": "2021-03-26",
    "detail": true
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "res_get_reports",
  "payload": [
```

Message Schema

- Request
- Response
- Errors

Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
{
  "priority": -1,
  "customService": "simple-service",
  "conditions": [],
  "_id": "605d649b0ab6fc0041efe9c8",
  "title": "cafe-table-1 delivery",
  "job": "605d649b0ab6fc0041efe9c7",
  "task": "605c17b65d884aa3ef39a8e0",
  "worker": "605c17b50ab6fc0041efe9b8",
  "created_at": "2021-03-26T04:35:39.159Z",
  "assigned_at": "2021-03-26T04:35:43.033Z",
  "started_at": "2021-03-26T04:35:47.971Z",
  "finished_at": "2021-03-26T04:35:48.316Z",
  "instructions": [
    {
      "issuer_type": "user",
      "performer_type": "worker",
      "id": "1",
      "action": {
        "fixed": false,
        "args": [
          {
            "type": "string",
            "format": null,
            "value": "5f1a3ee9cc6e2a00120d17d7",
            "domain": [],
            "default": null,
            "properties": [],
            "key": "destination",
            "value_alias": {
              "alias": "cafe_table_1",
              "value": "5f1a3ee9cc6e2a00120d17d7"
            }
          },
          {
            "type": "boolean",
            "format": null,
            "value": true,
            "domain": [],
            "default": null,
            "properties": [],
            "key": "goal_finishing",
            "value_alias": {
              "alias": "true",
              "value": true
            }
          }
        ],
        "options": {
          "user_input": false
        }
      },
      "options": {
        "user_input": false
      }
    },
    {
      "type": "number",
      "format": null,
      "value": 30,
      "domain": [],
      "default": null,
      "properties": [],
      "key": "finishing_timeout",
```


Message Schema

- Request
- Response
- Errors

Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
        "value_alias": {
          "alias": "30s",
          "value": 30
        },
        "options": {
          "max": 300,
          "min": 0,
          "user_input": true
        }
      },
      "func_name": "move"
    },
    "next": null,
    "X-Inst-Id": "c60ebd25bfd3efc2585ba88a16ad7401"
  }
],
"status": "finished",
"updated_at": "2021-03-26T04:35:48.322Z",
"result": {
  "status": "success",
  "status_msg": "just finished",
  "current": null,
  "returns": [
    {
      "id": "1",
      "X-Inst-Id": "c60ebd25bfd3efc2585ba88a16ad7401",
      "status": "success",
      "status_msg": "just finished",
      "started_at": "2021-03-26T04:35:47.971Z",
      "finished_at": "2021-03-26T04:35:48.316Z",
      "updated_at": "2021-03-26T04:35:48.322Z"
    }
  ],
  "ownership_executionTime": 0.038
},
"id": "605d649b0ab6fc0041efe9c8"
}
}
```

req_get_remain_reports

- **description:** Look up the list of reports that have not yet been completed by the Worker.
- **return:** [Report](#) list is returned.

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ detail	boolean	default is false, Recommended use false

Example of payload

Message Schema

Request

Response

Errors

Terminology

Action

Task

Worker

Report

Request Methods

req_login

req_login_api

req_get_preset

req_get_presets

req_get_report

req_get_reports

req_get_remain_reports

req_exec_preset

req_cancel_report

req_get_task

req_get_task_worker

req_compose_job

req_event_trigger

req_subscribe_worker

req_get_recent_failed

req_run_recent_failed

Server-sent events

feedback_report

feedback_instruction

feedback_instruction_group

feedback_worker_status

request

```
{
  "uuid": "6232c16e-98ad-477b-83a1-acb777d385dd",
  "method": "req_get_remain_reports"
}
```

response

```
{
  "isSuccess": true,
  "uuid": "6232c16e-98ad-477b-83a1-acb777d385dd",
  "method": "res_get_remain_reports",
  "payload": [
    {
      "id": "605d746f0ab6fc0041efe9ce",
      "title": "cafe-table-1 delivery",
      "worker": "605c17b50ab6fc0041efe9b8",
      "status": "running",
      "created_at": "2021-03-26T05:43:11.045Z",
      "assigned_at": "2021-03-26T05:43:13.226Z",
      "started_at": "2021-03-26T05:43:18.207Z",
      "finished_at": null,
      "updated_at": "2021-03-26T05:43:18.222Z"
    }
  ]
}
```

req_exec_preset

- **description:** Run a specific preset to generate a report that the Worker can perform.
- **return:** The id of the generated [Report](#) is returned.

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ id	string	preset's id

Example of payload

request

```
{
  "uuid": "26e935cb-9a08-4548-8d14-58da9c2beaf2",
  "method": "req_exec_preset",
  "params": {
    "id": "605c186b3ba1fe0041af0974"
  }
}
```

response

Message Schema

- Request
- Response
- Errors

Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
{
  "isSuccess": true,
  "uuid": "26e935cb-9a08-4548-8d14-58da9c2beaf2",
  "method": "res_exec_preset",
  "payload": {
    "id": "605c35890ab6fc0041efe9ba"
  }
}
```

req_cancel_report

- description:** Cancel a specific report
- return:** Information from a particular report is returned. [Report](#)

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ id	string	report's id
└ message	message	Reason for Cancellation
└ detail	boolean	default is false, Recommended use false

Example of payload

request

```
{
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "req_cancel_report",
  "params": {
    "id": "60f7d1f76cea710041ad9f52",
    "message": "'cancel from ws-open-api",
    "detail": true
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "70cb5d04-65c7-4314-b978-f2389906a28c",
  "method": "res_cancel_report",
  "payload": {
    "id": "60f7d5fa6cea710041ad9f61",
    "title": "New Job : 2021-07-21 5:08:16 pm",
    "worker": "60c02561d5bccb002b9c68b4",
    "status": "finished",
    "created_at": "2021-07-21T08:08:26.451Z",
    "assigned_at": "2021-07-21T08:08:27.402Z",
    "started_at": "2021-07-21T08:08:27.571Z",
    "finished_at": "2021-07-21T08:09:07.711Z",
    "updated_at": "2021-07-21T08:09:07.712Z"
  }
}
```

Message Schema

- Request
- Response
- Errors

Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report

- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
}  
}
```

req_get_task

- description:** Look up the Task of the Worker registered with the Concert.
- return:** Return the [Task](#).

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ name	string	task's name

Example of payload

request

```
{  
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",  
  "method": "req_get_task",  
  "params": {  
    "name": "robot-cafe-delivery"  
  }  
}
```

response

```
{  
  "isSuccess": true,  
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",  
  "method": "res_get_task",  
  "payload": {  
    "_id": "605c17b65d884aa3ef39a8e0",  
    "actions_hash": "e5661eeb66d0e755505e5e87c46006793f40edc8915b91c1f61f3c8413d8d56e",  
    "name": "robot-cafe-delivery",  
    "__v": 0,  
    "actions": [  
      {  
        "fixed": false,  
        "args": [  
          {  
            "type": "string",  
            "format": null,  
            "value": null,  
            "domain": [  
              {  
                "alias": "cafe_table_1",  
                "value": "5f1a3ee9cc6e2a00120d17d7"  
              },  
              {  
                "alias": "cafe_table_2",  
                "value": "5f1a3ef4cc6e2a00120d17d8"  
              }  
            ]  
          }  
        ]  
      }  
    ]  
  }  
}
```

Message Schema

- Request
- Response
- Errors

Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
{
  "alias": "1st_hall_1",
  "value": "5f1a415acc6e2a00120d17f8"
},
{
  "alias": "1st_hall_2",
  "value": "5f1a4165cc6e2a00120d17f9"
}
],
"default": {
  "alias": "cafe_table_1",
  "value": "5f1a3ee9cc6e2a00120d17d7"
},
"properties": [],
"key": "destination",
"options": {
  "user_input": false
}
},
{
  "type": "boolean",
  "format": null,
  "value": null,
  "domain": [
    {
      "alias": "true",
      "value": true
    },
    {
      "alias": "false",
      "value": false
    }
  ],
"default": {
  "alias": "true",
  "value": true
},
"properties": [],
"key": "goal_finishing",
"options": {
  "user_input": false
}
},
{
  "type": "number",
  "format": null,
  "value": null,
  "domain": [
    {
      "alias": "0s",
      "value": 0
    },
    {
      "alias": "10s",
      "value": 10
    },
    {
      "alias": "20s",
      "value": 20
    }
  ],
}
```

Message Schema

Request
Response
Errors

Terminology

Action
Task
Worker
Report

Request Methods

req_login
req_login_api
req_get_preset
req_get_presets
req_get_report
req_get_reports
req_get_remain_reports
req_exec_preset
req_cancel_report

req_get_task
req_get_task_worker
req_compose_job
req_event_trigger
req_subscribe_worker
req_get_recent_failed
req_run_recent_failed

Server-sent events

feedback_report
feedback_instruction
feedback_instruction_group
feedback_worker_status

```
      "alias": "30s",
      "value": 30
    },
    {
      "alias": "40s",
      "value": 40
    },
    {
      "alias": "50s",
      "value": 50
    }
  ],
  "default": {
    "alias": "30s",
    "value": 30
  },
  "properties": [],
  "key": "finishing_timeout",
  "options": {
    "max": 300,
    "min": 0,
    "user_input": true
  }
},
{
  "func_name": "move"
},
{
  "fixed": false,
  "args": [
    {
      "type": "string",
      "format": null,
      "value": null,
      "domain": [
        {
          "alias": "1st_charging_station",
          "value": "605c16738d3e8c0040b3adfb"
        }
      ],
      "default": {
        "alias": "1st_charging_station",
        "value": "605c16738d3e8c0040b3adfb"
      },
      "properties": [],
      "key": "station",
      "options": {
        "user_input": false
      }
    }
  ],
  "func_name": "dock"
}
],
"created_at": "2021-03-25T04:55:18.621Z",
"description": "",
"updated_at": "2021-03-25T04:55:18.621Z",
"id": "605c17b65d884aa3ef39a8e0"
}
}
```

Message Schema

- Request
- Response
- Errors

Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

req_get_task_worker

- description:** Look up a Worker who can perform a Task registered with Concert.
- return:** Returns the [Worker](#) list.

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ name	string	task's name

Example of payload

request

```
{
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "req_get_task_worker",
  "params": {
    "name": "robot-cafe-delivery"
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "res_get_task_worker",
  "payload": [
    {
      "type_specific": {
        "battery": {
          "battery_level": 75,
          "charging_status": 0
        },
        "location": {
          "map": "5f19503fdbb58b001973eca1",
          "pose2d": {
            "x": 1.8998796147672543,
            "y": 3.9563125000000003,
            "theta": 0
          },
          "semantic_location": null
        },
        "ip": "127.0.0.1"
      },
      "id": "605c17b50ab6fc0041efe9b8",
      "name": "simple-worker",
      "status": "idle",
      "created_at": "2021-03-25T04:55:17.665Z",
      "updated_at": "2021-03-30T03:01:32.592Z"
    }
  ]
}
```

Message Schema

- Request
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- Errors

Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

req_compose_job

- description:** Dynamically configure the Job using Actions in the Task.
- return:** The id of the generated [Report](#) is returned.

Sample Task Schema

```
{
  "name" : "simple-delivery",
  "actions" : [
    {
      "func_name": "move",
      "args": [
        {"key": "destination", "type": "string"},
        {"key": "timeout", "type": "number"}]
    },
    {
      "func_name": "dock",
      "args": [{"key": "station", "type": "string"}]
    }
  ]
}
```

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ title	string	
└ worker	string	optional, worker 's id
└ task	string	task's name
└ instructions	Array	Lists the features that the Worker (=robot) will run. Report.instructions
└ (Anonymous)	object	eg) { "func_name": "move", "args":{"key": "destination", "type": "string"},{"key": "timeout", "type": "number"}}}
└ id	string	Important, this value used by "Server-sent event:feedback-instruction"
└ func_name	string	eg) { "func_name": "move" } -> "move"
└ args	object	eg) {"args":{"key": "destination", "type": "string"},{"key": "timeout", "type": "number"}}-> { "destination": "1st_hall_1", "timeout": 30 } for more informations about destinations. location lookup API: http://concertlp:10602/configuration/api/Locations?type=waypoint

Example of payload

request

```
{
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "req_compose_job",
  "params": {
    "title": "move to cafe_table_1 & go to charging",
    "task": "robot-cafe-delivery",
    "instructions": [
      {
        "id": "1",
        "func_name": "move",
```


Message Schema

- Request
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- Terminology
- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
      "args": {
        "destination": "5f1a3ee9cc6e2a00120d17d7",
        "goal_finishing": false,
        "finishing_timeout": null
      }
    },
    {
      "id": "2",
      "func_name": "dock",
      "args": {
        "station": "605c16738d3e8c0040b3adfb"
      }
    }
  ]
}
}
```

```
{
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "req_compose_job",
  "params": {
    "title": "move to cafe_table_1 & wait(3 sec) & move to 1st_hall_2 & wait for event-trigger(10sec)& move to cafe_table_1",
    "task": "Virtual Delivery",
    "instructions": [
      { "id": "1", "func_name": "move", "args": {"destination": "5f1a3ee9cc6e2a00120d17d7"}},
      { "id": "2", "func_name": "wait", "args": {"type": "duration", "timeout": 3000}},
      { "id": "3", "func_name": "move", "args": {"destination": "5f1a4165cc6e2a00120d17f9"}},
      { "id": "4", "func_name": "event_listener", "args": {"timeout": "10"}},
      { "id": "5", "func_name": "move", "args": {"destination": "5f1a3ee9cc6e2a00120d17d7"}}
    ]
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "res_compose_job",
  "payload": {
    "id": "6061877e0ab6fc0041efe9da"
  }
}
```

req_event_trigger

- **description:** If the instructions created by [compose-job](#) include event_listener, it can be completed by sending the instruction id.

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└─ reportId	string	
└─ instId	string	

Message Schema

Request
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Errors

Terminology

Action
Task
Worker
Report

Request Methods

req_login
req_login_api
req_get_preset
req_get_presets
req_get_report
req_get_reports
req_get_remain_reports
req_exec_preset
req_cancel_report
req_get_task
req_get_task_worker
req_compose_job
req_event_trigger
req_subscribe_worker
req_get_recent_failed
req_run_recent_failed

Server-sent events

feedback_report
feedback_instruction
feedback_instruction_group
feedback_worker_status

Example of payload

request

```
{
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "req_event_trigger",
  "params": {
    "reportId": "6164f16650de88733e4aa30c",
    "instId": "2"
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "res_event_trigger"
}
```

req_subscribe_worker

- **description:**subscribe [feedback-worker-status](#)

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ enable	boolean	true/false = subscribe/unsubscribe
└ interval	number	seconds

Example of payload

request

```
{
  "uuid": "2a65e329-76cb-4df6-ae20-83d515aa641f",
  "method": "req_subscribe_worker",
  "params": {
    "enable": true,
    "interval": 10
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "2a65e329-76cb-4df6-ae20-83d515aa641f",
  "method": "res_subscribe_worker"
}
```

Message Schema

- Request
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Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

req_get_recent_failed

- description:** Check if there are any recent failed reports that can be retried
- return:** The id of the generated [Report](#) is returned.

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ worker	string	required, worker 's id

Example of payload

request

```
{
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "req_get_recent_failed",
  "params": {
    "worker": "62a1af8973a42fe7224e34eb"
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "res_get_recent_failed",
  "payload": {
    "id": "62a30551e3f734002b975fd4"
  }
}
```

req_run_recent_failed

- description:** Create a report of the retried job which is starting on the last failed instruction.
- return:** The id of the generated [Report](#) is returned.

Schema

Name	Type	Description
uuid	string	
method	string	
params	object	
└ worker	string	required, worker 's id

Example of payload

request

Message Schema

Request

Response

Errors

Terminology

Action

Task

Worker

Report

Request Methods

req_login

req_login_api

req_get_preset

req_get_presets

req_get_report

req_get_reports

req_get_remain_reports

req_exec_preset

req_cancel_report

req_get_task

req_get_task_worker

req_compose_job

req_event_trigger

req_subscribe_worker

req_get_recent_failed

req_run_recent_failed

Server-sent events

feedback_report

feedback_instruction

feedback_instruction_group

feedback_worker_status

```
{
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "req_run_recent_failed",
  "params": {
    "worker": "62a1af8973a42fe7224e34eb"
  }
}
```

response

```
{
  "isSuccess": true,
  "uuid": "61983e7a-f2bc-49e1-8fe5-83cdb41fecf3",
  "method": "res_run_recent_failed",
  "payload": {
    "id": "62a3138352f8c49a1157284b"
  }
}
```

Server-sent events

feedback-report

- **description:** Feedback event for the progress of the report
- **return:** Send status information for the report you are performing.

Schema

Name	Type	Description
uuid	string	
method	string	
payload	object	
└ id	string	report's id
└ title	string	report's title
└ worker	string	Worker's id performing the Report
└ created_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ The time when the report was generated
└ assigned_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZThe time when the Report was assigned to Worker
└ started_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ The time when the Worker started performing the Report.
└ finished_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ The time when the Report has finished
└ status	string	Report.status

Example of payload

Report was created

```
{
  "uuid": "4197dcb0-faf2-4bc9-9ac8-1c18f78fe62a",
  "method": "feedback-report",
  "payload": {
    "id": "6061b08d0ab6fc0041efe9e4",
  }
}
```

Message Schema

- Request
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Terminology

- Action
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Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
{
  "title": "move to cafe_table_1 & go to charging",
  "worker": null,
  "created_at": "2021-03-29T10:48:45.371Z",
  "assigned_at": null,
  "started_at": null,
  "finished_at": null,
  "status": "pending"
}
```

Report was assigned(=Report taken by Worker)

```
{
  "uuid": "d4c738ce-57ac-4591-acb7-8b05e8756cf3",
  "method": "feedback-report",
  "payload": {
    "id": "6061b08d0ab6fc0041efe9e4",
    "title": "move to cafe_table_1 & go to charging",
    "worker": "605c17b50ab6fc0041efe9b8",
    "created_at": "2021-03-29T10:48:45.371Z",
    "assigned_at": "2021-03-29T10:48:47.153Z",
    "started_at": null,
    "finished_at": null,
    "status": "assigned"
  }
}
```

Report was running(=Worker will execute Report.instruction)

```
{
  "uuid": "192ce378-a2e8-4e1b-89e0-b70bcd92c129",
  "method": "feedback-report",
  "payload": {
    "id": "6061b08d0ab6fc0041efe9e4",
    "title": "move to cafe_table_1 & go to charging",
    "worker": "605c17b50ab6fc0041efe9b8",
    "created_at": "2021-03-29T10:48:45.371Z",
    "assigned_at": "2021-03-29T10:48:47.153Z",
    "started_at": "2021-03-29T10:48:52.078Z",
    "finished_at": null,
    "status": "running"
  }
}
```

Report was finished

```
{
  "uuid": "ed30a250-4ca1-4c29-abda-431fb86b144c",
  "method": "feedback-report",
  "payload": {
    "id": "6061b08d0ab6fc0041efe9e4",
    "title": "move to cafe_table_1 & go to charging",
    "worker": "605c17b50ab6fc0041efe9b8",
    "created_at": "2021-03-29T10:48:45.371Z",
    "assigned_at": "2021-03-29T10:48:47.153Z",
    "started_at": "2021-03-29T10:48:52.078Z",
    "finished_at": "2021-03-29T10:49:04.553Z",
    "status": "finished"
  }
}
```

Message Schema

- Request
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Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

feedback-instruction

- **description:**
 - Feedback event for the Instruction(=user defined instruction) progress.
 - **purpose of feedback-instruction:** to check the status of user defined instruction
- **return:** Sends status information for the performing Report.instruction.

Schema

Name	Type	Description
uuid	string	
method	string	
payload	object	
└ id	string	instruction's id(=user defined instruction's id)
└ reportId	string	report's id
└ workerId	string	worker's id
└ status	string	Instruction.status
└ started_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
└ finished_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
└ updated_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
└ detail	string	instruction's detail

Example of payload

Feedback on the started Instructions.

```
{
  "uuid": "9ae33244-75bd-47ec-8b08-70c7c6baba53",
  "method": "feedback-instruction",
  "payload": {
    "reportId": "61dcd8f70d1f7a00427ce877",
    "workerId": "61db8db60d1f7a00427ce7c9",
    "started_at": "2022-01-11T01:10:43.278Z",
    "finished_at": null,
    "updated_at": "2022-01-11T01:10:43.291Z",
    "status": "started",
    "id": "7d3f47a717625d5a80fd098f11ac3ad6",
    "detail": {
      "id": "14",
      "action": {
        "fixed": false,
        "args": [
          {
            "type": "string",
            "value": "5f1a4165cc6e2a00120d17f9",
            "domain": [],
            "default": null,
            "properties": [],
            "key": "destination",
            "value_alias": {
              "value": "5f1a4165cc6e2a00120d17f9",
              "alias": "1st_hall_2"
            }
          }
        ]
      }
    }
  },
}
```

Message Schema

- Request
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Terminology

- Action
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- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
{
  "type": "number",
  "value": -1,
  "domain": [],
  "default": null,
  "properties": [],
  "key": "finishing_timeout",
  "value_alias": {
    "alias": "-1",
    "value": -1
  }
},
{
  "func_name": "move"
},
{
  "next": null,
  "X-Inst-Id": "7d3f47a717625d5a80fd098f11ac3ad6",
  "issuer_type": "user",
  "performer_type": "worker"
}
}
```

Feedback on the finished Instruction.

```
{
  "uuid": "8fe5fe01-9c0a-4665-9748-266a2c907a21",
  "method": "feedback-instruction",
  "payload": {
    "reportId": "61dcd8f70d1f7a00427ce877",
    "workerId": "61db8db60d1f7a00427ce7c9",
    "started_at": "2022-01-11T01:10:43.278Z",
    "finished_at": "2022-01-11T01:10:44.972Z",
    "updated_at": "2022-01-11T01:10:44.982Z",
    "status": "done",
    "id": "7d3f47a717625d5a80fd098f11ac3ad6",
    "detail": {
      "id": "14",
      "action": {
        "fixed": false,
        "args": [
          {
            "type": "string",
            "value": "5f1a4165cc6e2a00120d17f9",
            "domain": [],
            "default": null,
            "properties": [],
            "key": "destination",
            "value_alias": {
              "value": "5f1a4165cc6e2a00120d17f9",
              "alias": "1st_hall_2"
            }
          }
        ]
      }
    }
  },
  {
    "type": "number",
    "value": -1,
    "domain": [],
    "default": null,
    "properties": [],
    "key": "finishing_timeout",
    "value_alias": {
```

message Schema

- Request
- Response
- Errors
- Terminology
- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
      "alias": "-1",
      "value": -1
    }
  ],
  "func_name": "move"
},
"next": null,
"X-Inst-Id": "7d3f47a717625d5a80fd098f11ac3ad6",
"issuer_type": "user",
"performer_type": "worker"
}
}
```

feedback-instruction-group

- description:
 - Feedback event for the Whole Instruction progress
 - purpose of feedback-instruction-group:** Additional instructions are added to user defined instruction by the concert-system. For example, assume that a robot is instructed with **Move to Location A** by the user, which is located after passing through an automatic door, robot will need to use the automatic door before moving to Location A. The overall process (**Instruction group**) of moving to Location A will be expanded by the concert-system like this [**use_autodoor, Move To Location A(X-Instruction=user defined instruction)**]
- return: Sends status information for the performing Report.instruction.

Schema

Name	Type	Description
uuid	string	
method	string	
payload	object	
└ id	string	instruction's id(=generated instruction's id by Concert-system)
└ reportId	string	report's id
└ workerId	string	worker's id
└ status	string	Instruction.status
└ started_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
└ finished_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
└ updated_at	string	format ISO 8601, YYYY-MM-DDTHH:mm:ss.sssZ
└ X-Inst-Info	object	
└└ X-Inst-Id	string	X-Instruction's id(=user defined instruction's id)
└└ seq	string	X-Inst-Group sequence
└└ total	string	count in X-Inst-Group
└ detail	string	instruction's detail

Example of payload

Feedback on the started Instructions.

```
{
  "uuid": "27d3ad09-27ac-40db-a69e-dfa5a701d024",
  "method": "feedback-instruction-group",
```


Message Schema

- Request
- Response
- Errors

Terminology

- Action
- Task
- Worker
- Report

Request Methods

- req_login
- req_login_api
- req_get_preset
- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
"payload": {
  "reportId": "61dcd8f70d1f7a00427ce877",
  "workerId": "61db8db60d1f7a00427ce7c9",
  "started_at": "2022-01-11T01:10:31.796Z",
  "finished_at": null,
  "updated_at": "2022-01-11T01:10:31.805Z",
  "id": "10",
  "status": "started",
  "X-Inst-Info": {
    "X-Inst-Id": "7d3f47a717625d5a80fd098f11ac3ad6",
    "seq": 1,
    "total": 5
  },
  "detail": {
  }
}
```

Feedback on the finished Instruction.

```
{
  "uuid": "760586bb-1090-448a-9277-722ff2a05c6f",
  "method": "feedback-instruction-group",
  "payload": {
    "reportId": "61dcd8f70d1f7a00427ce877",
    "workerId": "61db8db60d1f7a00427ce7c9",
    "started_at": "2022-01-11T01:10:31.796Z",
    "finished_at": "2022-01-11T01:10:37.999Z",
    "updated_at": "2022-01-11T01:10:38.011Z",
    "id": "10",
    "status": "done",
    "X-Inst-Info": {
      "X-Inst-Id": "7d3f47a717625d5a80fd098f11ac3ad6",
      "seq": 1,
      "total": 5
    }
  }
}
```

feedback-worker-status

- **description:**
 - *Feedback event for worker's status*
- **return:** *Sends status information for the worker*

Schema

Name	Type	Description
uuid	string	
method	string	
payload	object	
└ uuid	string	<i>worker's uuid</i>
└ name	string	
└ status	string	Worker.status
└ status	string	Instruction.status

Message Schema

Request

Response

Errors

Terminology

Action

Task

Worker

Report

Request Methods

req_login

req_login_api

req_get_preset

req_get_presets

req_get_report

req_get_reports

req_get_remain_reports

req_exec_preset

req_cancel_report

req_get_task

req_get_task_worker

req_compose_job

req_event_trigger

req_subscribe_worker

req_get_recent_failed

req_run_recent_failed

Server-sent events

feedback_report

feedback_instruction

feedback_instruction_group

feedback_worker_status

Name	Type	Description
└ type_specific	object	Worker's own properties
└ location	object	worker's current position of map
└ map	string	map's id
└ pose2d	object	
└ x	number	x-coordinate
└ y	number	y-coordinate
└ theta	number	radian
└ battery	object	battery status object
└ battery_level	number	percentage of battery
└ now_charging	boolean	
└ home_station	object	worker's charging-station, specified robot's configuration, used by auto-charging policy
└ name	string	charging-station's name
└ id	string	charging-station's id
└ ip	string	

Example of payload

```
{
  "uuid": "c1ee0d5a-3d5f-4d63-9802-93f137d96170",
  "method": "feedback-worker-status",
  "payload": {
    "uuid": "vw_dy_robot",
    "name": "vw_dy_robot",
    "status": "idle",
    "type_specific": {
      "battery": {
        "battery_level": 0,
        "now_charging": false
      },
      "location": {
        "map": "5f19503fdbb58b001973eca1",
        "pose2d": {
          "x": 4.1573822356017835,
          "y": 23.2007082973512,
          "theta": 4.635616856178602
        }
      }
    },
    "ip": "127.0.0.1",
    "home_station": {
      "name": "Docking Station",
      "id": "5f6994a71021920034fa6317"
    }
  }
}
```

Feedback on the finished Instruction.

```
{
  "uuid": "760586bb-1090-448a-9277-722ff2a05c6f",
  "method": "feedback-instruction-group",
```

Message Schema

- Request
- Response
- Errors

Terminology

- Action
- Task
- Worker
- Report

Request Methods

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- req_get_presets
- req_get_report
- req_get_reports
- req_get_remain_reports
- req_exec_preset
- req_cancel_report
- req_get_task
- req_get_task_worker
- req_compose_job
- req_event_trigger
- req_subscribe_worker
- req_get_recent_failed
- req_run_recent_failed

Server-sent events

- feedback_report
- feedback_instruction
- feedback_instruction_group
- feedback_worker_status

```
"payload": {
  "reportId": "61dcd8f70d1f7a00427ce877",
  "workerId": "61db8db60d1f7a00427ce7c9",
  "started_at": "2022-01-11T01:10:31.796Z",
  "finished_at": "2022-01-11T01:10:37.999Z",
  "updated_at": "2022-01-11T01:10:38.011Z",
  "id": "10",
  "status": "done",
  "X-Inst-Info": {
    "X-Inst-Id": "7d3f47a717625d5a80fd098f11ac3ad6",
    "seq": 1,
    "total": 5
  }
}
```